

PITE-RONNSKAR, Sweden

WMO: 021760

Lat: 65.034N

Lon: 21.561E

Elev: 5

StdP: 101.27

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.8	-20.4	-27.6	0.3	-20.9	-23.4	0.5	-20.1	15.8	-2.1	14.3	-5.0	2.6	220	0.770

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	21.3	17.6	19.8	16.3	18.6	15.5	18.6	20.6	17.3	19.1	16.3	18.0	4.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.7	19.9	16.5	11.8	18.6	15.5	11.0	17.5	52.5	20.5	48.8	19.1	45.8	18.0	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	12.2	10.7	-26.6	24.0	5.9	1.9	-30.9	25.4	-34.3	26.5	-37.7	27.6	-42.0	29.0		
(5)			-26.6	19.0	5.5	1.4	-30.6	20.0	-33.8	20.9	-36.9	21.7	-40.9	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-6.8	-8.2	-4.2	0.7	6.4	11.8	15.6	14.4	10.0	4.3	-0.5	-3.9	(6)
(7)		DBStd	9.05	6.54	7.03	4.95	3.01	3.22	2.75	2.43	2.43	2.74	3.43	4.33	5.59	(7)
(8)		HDD10.0	2837	519	510	439	278	121	13	0	1	34	177	314	430	(8)
(9)		HDD18.3	5470	778	743	698	528	371	197	92	124	251	435	564	688	(9)
(10)		CDD10.0	417	0	0	0	0	8	66	173	137	33	1	0	0	(10)
(11)		CDD18.3	9	0	0	0	0	0	0	7	2	0	0	0	0	(11)
(12)		CDH23.3	14	0	0	0	0	1	0	10	3	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.0	5.4	5.0	5.0	4.4	4.5	4.4	4.7	5.0	5.4	5.7	5.7	(14)	
(15)	Precipitation	PrecAvg	507	34	24	21	27	34	50	64	58	52	50	37	(15)	
(16)		PrecMax	679	72	49	50	56	90	89	159	110	158	155	97	98	(16)
(17)		PrecMin	344	6	5	6	6	5	9	23	11	10	9	12	12	(17)
(18)		PrecStd	97	15	11	12	13	20	20	30	31	34	31	24	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	6.9	8.5	12.0	18.3	20.1	24.2	22.2	17.7	12.4	8.8	6.1	(19)
(20)			MCWB	3.5	3.9	4.3	6.1	11.0	13.5	19.5	18.4	12.3	8.8	6.3	4.1	(20)
(21)		2%	DB	3.3	4.2	6.2	8.9	14.6	18.4	22.3	20.4	15.8	10.6	6.7	4.5	(21)
(22)			MCWB	1.5	1.8	2.7	4.3	9.2	12.8	18.7	17.2	12.5	8.9	5.7	3.1	(22)
(23)		5%	DB	1.7	1.9	4.1	6.4	12.4	17.0	20.7	19.1	14.7	9.8	5.9	3.5	(23)
(24)			MCWB	0.7	0.4	1.3	3.2	8.3	12.7	17.3	16.2	12.6	8.7	5.1	2.4	(24)
(25)		10%	DB	0.7	0.4	2.0	4.5	10.7	15.8	19.3	18.1	13.9	9.0	4.9	2.6	(25)
(26)			MCWB	0.0	-0.4	0.3	2.3	7.4	12.4	16.4	15.6	12.2	7.8	4.1	1.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	4.1	4.8	6.6	12.2	16.0	20.9	19.2	14.7	10.3	6.7	4.5	(27)
(28)			MDCB	5.8	6.6	8.1	10.7	16.1	17.7	23.0	21.3	15.4	10.7	7.5	5.3	(28)
(29)		2%	WB	1.8	2.1	3.0	4.9	10.2	14.6	19.5	18.0	13.8	9.6	6.1	3.5	(29)
(30)			MDCB	2.7	4.0	5.6	8.0	13.5	16.6	21.6	19.8	14.5	10.2	6.7	4.3	(30)
(31)		5%	WB	0.9	0.6	1.6	3.6	8.9	13.8	18.1	17.0	13.3	8.8	5.2	2.7	(31)
(32)			MDCB	1.6	1.5	3.4	5.9	11.7	15.9	20.0	18.5	14.1	9.7	5.8	3.3	(32)
(33)		10%	WB	0.1	-0.3	0.6	2.6	7.7	12.9	16.9	16.2	12.6	7.8	4.3	1.9	(33)
(34)			MDCB	0.7	0.4	1.7	4.2	10.2	15.0	18.8	17.6	13.6	8.8	4.8	2.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	6.5	6.6	5.2	4.8	4.3	4.2	4.3	4.7	4.1	3.8	4.7	(35)
(36)			MCDBR	4.9	6.9	8.8	8.6	8.4	6.4	6.0	5.6	5.7	4.5	3.7	4.0	(36)
(37)			MCWBR	4.4	5.8	6.3	5.3	4.8	3.6	3.8	3.9	3.8	3.6	3.2	3.6	(37)
(38)		5% WB	MCDBR	4.9	6.5	7.7	7.9	7.5	4.5	4.6	4.3	4.0	3.6	3.4	3.7	(38)
(39)			MCWBR	4.5	5.6	5.8	5.1	4.6	3.1	3.5	3.3	3.3	3.2	3.2	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	N/A	0.275	0.303	0.337	0.344	0.348	0.363	0.350	0.321	0.302	0.339	0.348	(40)	
(41)		taud	N/A	2.145	2.204	2.274	2.449	2.495	2.472	2.501	2.556	2.389	2.598	2.620	(41)	
(42)		Ebn at Noon	N/A	608	767	830	865	868	840	818	770	593	287	120	(42)	
(43)		Edh at Noon	N/A	57	85	102	95	93	93	82	63	46	21	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.11	0.69	2.20	3.91	4.95	5.47	5.05	3.78	2.18	0.83	0.18	0.03	(44)	
(45)		RadStd	0.01	0.12	0.21	0.30	0.49	0.44	0.47	0.35	0.29	0.15	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page